Acoustic Recordings from Antiquity

Abstract—Pioneering experiments establishing the principles of recalling ancient sounds from antiquity are reported.

Widespread research on recalling from the past actual sounds, voices, music, etc., adventitiously recorded by ancient peoples (or events) upon the "surface" or within the substance of a wide variety of objects and artifacts crafted (or evolved) from "plastic" media warrants intensified efforts at the present time, because of recent developments in electronic signal analysis which can ferret out "signals" buried in "noise".

This letter is primarily intended to call attention to the potentials of Acoustic Archaeology and to record the early experiments which established the principle.

Two areas of the author's investigation, which began in 1961, will be of interest: 1) the recording of sound on wheel-thrown clay pots, and 2) the recording of sound in paint strokes applied to canvas.

The sound-reproducing system used consisted of a crystal cartridge (Asiatic Corp., Model 2) such as is used in phono pickups. The cartridge was connected directly to a set of inexpensive earphones (Trimm "Acme." 2000 ohms). The chuck of cartridge could be fitted with "needles" of any suitable material, length, shape, etc. In all instances the cartridge was held in the fingers and could be positioned against a revolving pot mounted on a phono turntable (adjustable speed) or "stroked" along a paint stroke, etc.

Sound Recorded on Pottery

First Example

This consisted of a pot of fine clay, hand thrown on a potter's wheel. The wheel in this example was an old, student-made wheel, constructed of an automobile crankshaft and flywheel mounted in a (too) light wooden frame. Persistently out of alignment, the wheel had a noisy vibration almost amounting to a chatter. The pot produced on this wheel was fired at low temperatures.

When the pot was suitably mounted on the phono turntable and against the side of the revolving pot was held the phono cartridge (fitted, in this instance, with a "needle" consisting of a flat-ended sliver of wood three-quarters of an inch long) the low-frequency chatter could be heard in the earphones.

Second Example

This was similar to the first example except it was a commercial pot which had been hand thrown on a motor-driven potter's wheel. The 60 Hz motor was mounted directly on the frame supporting the wheel to which it imparted a loud hum. Using the phono cartridge as above with a similar needle, areas could be found on the surface of the revolving pot where the hum could be picked up.

In both examples, it should be noted, the last act of the potter, prior to removing the pot from the wheel, was to "smooth" the surface of the pot using the "sharp" edge of a thin rib of wood.

Sound Recorded in Paint Strokes

This is of particular interest as it introduces the possibility of actually recalling and hearing the voices and words of eminent personages as recorded in the paint of their portraits or of famous artists in their pictures.

First Example

A canvas affixed to a small, square wooden frame was so stationed as to be able to vibrate freely. This it would do when "spoken to" or where subjected to music from a nearby phonograph—as determined by touching it to the "needle" (wooded sliver) of the crystal cartridge and listening in the earphones.

With an artist's brush, paint strokes were applied to the surface of the canvas using "oil" paints involving a variety of plasticities, thicknesses, layers, etc., while martial music was played on the nearby phonograph. Visual examination at low magnification showed that certain strokes had the expected transverse striated appearance. When such strokes, after drying, were gently stroked by the "needle" (small, wooden, spade-like) of the crystal cartridge, at as close to the original stroke speed as possible, short snatches of the original music could be identified.

Second Example

This is to record the finding of a spoken word in an oil portrait. The word was "blue" and was located in a blue paint stroke—as if the artist was talking to himself or to the subject. Parenthetically, the search was long and

tedious. The principle, however, was established.

Many situations leading to the possibility of adventitious acoustic recording in past times have been given consideration. These, for example, might consist of scratches, markings, engravings, grooves, chasings, smears, etc., on or in "plastic" materials encompassing metal, wax, wood, bone, mud, paints, crystal, and many others. Artifacts could include objects of personal adornment, sword blades, arrow shafts, pots, engraving plates, paintings, and various items of calligraphic interest.
It is believed that this is the first public disclosure in this interesting field.

Richard G. Woodbridge, III
Box 111
Princeton Junction, N. J. 08550

I don't know what to think about this. It could be engineer humor, though it wasn't published in the April issue. "
http://tenser.typepad.com/tenser_said_the_tensor/2006/02/pottery_recordi.html
(end quote from 'Tensor' at above link, visit the link for more – ed.)

The basic idea is to estimate the speed at which the pot would have been spun on the potters wheel, and scan a laser across it a that speed, analyzing the reflected light patterns on a computer. Another technique is to drag a light mechanical stylus like a bit of straw or a wooden sliver across it at about the speed used to spin the pot. They said the surface of the pottery was audio modulated by ambient sounds, using the potters hand as the stylus. The same thing has been postulated about paint strokes.
Regardless of the veracity of the Belgian claim, the concept illustrates how an orderly pattern can be created and held within a pocket of 'relative randomness'.

For the purposes of its use as a pot, the microscopic surface texture ridges on the pottery are so small as to appear smooth, and therefore we consider them to be only random fluctuations because they are below the 'zoom' level we are looking at. But if we 'zoom' in and actually look at them they are not truly random. They in fact carry data.
Data = Organization.

All randomness is relative, in that it is only random within parameters. Let us imagine we have a 1 foot square box full of ping pong balls. Each ping pong ball has a mark somewhere on its surface. So we fill up the box with the ping pong balls and shake it well. The balls should be arranged randomly, right? The marks should point in all different random directions, Right?

Right, but only from one 'zoom' level of perception. They are only 'so' random. The placement of balls is not truly random. They are only random within parameters. What parameters? Dump the box on the floor.

Are they randomly placed now? Nope, they will still, by their order, show you where they hit the floor and from what angle, and how hard. But they are a lot more randomly placed than when they were in the box. Of course, they are still in the larger box formed by the room in which the floor exists. Then in the box formed by the building, etc on up the scale.

Getting a random interference sample is like filling a box with ping pong balls, so that we have a box of randomness, which we can work with in an orderly fashion. Write the word love on the side of it and shake it again. The balls will assume a pattern that most easily syncs with the QDS we introduced through our archetypical sample – more or less.

In order to get a really good sample, we want lots and lots of 'ping pong balls', and we want them to fit into a conveniently small package. We also need a method of reading the memory, other than carefully touching the side of the box so as to not disturb it.

Maybe we could mix a bunch of glue into the pig pong balls before we shake it, and then let the glue set. Now, our sample will stay the same (relatively) and we can pick it up and carry it around without destroying the data.

This same concept applies to the arrangement of atoms and / or molecules in physical matter, as it does to larger material bodies.

All we need to do is get a sufficient quantity of a material which is finely divided, composed of many small parts, and subject it to the energy signature that we want to capture while changing it in some way.

This type of sampling essentially uses the probability influencing of the QDS to create a glyphic sample, using the ping pong balls / molecules / crystals / atoms in a material sample.

Though it may be incomprehensibly complex, the possible alignment combinations of water molecules in a bucket of water is finite. It is a very high number, but not as high as the possible alignment of all water molecules in a swimming pool, for example.

And because of the laws of conservation of energy / a system wanting to 'right itself'… the alignment that the water molecules assume when disturbed will be that which is 'easiest' for them to maintain, and that will be defined by the ambient energy conditions which the water experiences (and those ambient conditions in turn are , I contend, influenced by the QDS).

Water is a passive write material in that it does not generally require a large amount of energy other than the QDS to pick up a sample. It will (metaphorically) take a 'photograph' in very low 'lighting conditions', but the 'photographs' fade rapidly unless the water is insulated in some way from outside influences.

This same tendency of water to easily pick up a QDS means that it is also easily corrupted over time. Water samples should be stored in glass, since the glass allows the QDS to pass through but insulates the water somewhat from outside energies. I can best describe the dynamics of shielding the water QDS data with glass as being similar to the trick of boiling water in a paper cup. Water samples should be stored in a cool dark quiet place or a secure box with thick walls.

Virtually any material will act as a passive write QDS recording media, but only a few materials like water have a broad response range in that will adapt easily to almost any QDS. For recording long, complex, abstract sequences of thought, a good method is to use water as the primary recording media, and then transfer the final sample from the water to a more permanent media to be used in a read only mode when broadcasting.

Metamorphic QDS sampling

Metamorphic sampling is one of the methods which I use most, and have spent considerable time developing. Metamorphic sampling has the advantages of adapting easily to nearly any QDS, being non-rewritable and thus much less prone to degrading over time, and allowing one to impress a QDS on a material compound that would not ordinarily be compatible with the QDS in a passive write mode. It allows you to permanently impress a 'blue' QDS on a 'red' material, metaphorically speaking.

The basic concept in metamorphic sampling is similar to the known and conventional method of determining the condition of the earth's magnetic field in the past by examining the molecular structure of rocks, so a brief description follows.
The earth's magnetic field exerts a subtle influence on the internal (supposedly random) structure of some types of rock, while the rock is forming or solidifying. Once the rock has formed and solidified, the internal structure remains much as it was, though the earth's magnetic field changes in the intervening years. The rocks can tell scientists about the condition of the earth's magnetic field when the rocks formed because the imprint which the earth's magnetic field left on the rock remains in the rock, though the field is now different than when the rocks formed. The rocks have in this case acted as metamorphic recording media for data concerning the earth's magnetic field. At one point in their history, the rocks were susceptible to the earth's magnetic field. Then, they went through a metamorphosis – they changed fundamentally and irreversibly into a different kind of rock, one which is not as susceptible to the earth's magnetic field. So, the impression that the earth's magnetic field left on the rocks was 'frozen' in time, and here for us to look at today.

In this document, we are dealing with Quantum Data Structures and probability fields, but the mechanics is similar.

Metamorphic QDS sampling through reduction

Imagine that you have a glass, and you smash it with a hammer, and then grind it down with a mortar and pestle into powder. What is the QDS of the glass now, and how does it differ from the QDS of the glass before it was reduced to powder? It differs in that where there was one glass, now there are many glasses. Where the original glass had a distinctive shape, the shape of the glass fragments is 'random'.

But if we look at the fragments, we will see two types of order, at least- one which is easy to see in conventional scientific fashion with appropriate equipment I am sure, and it is an order which tells us about the material characteristics of the glass itself. The particles will not be truly random in size; they will have a general shape which is determined by the physical properties of the glass specimen. They will tend to be sharp edged elongated splinters, elliptical chips, or broken fragments

thereof. They will all be unique variations on those themes, but they will for the most part fit that description. So they are not fully random.

The second and more esoteric order which emerges is that the QDS of the ambient surroundings when the glass was crushed will be reflected subtly in the supposedly random variant contours of the glass fragments, since the QDS acts as a probability field which influences the minute chance events occurring when the glass is fractured, forming tinier patterns on the edges of the already tiny splinters, influencing the exact surface contours of the chips … in the manner that relieves the most 'probability stress' and makes it 'easiest' for the glass to sync with the ambient energy field conditions in it's new form.

The third and more subtle QDS will be data concerning the history of the glass before it was crushed, unless the glass was annealed prior to crushing, and this QDS will be 'quieter' that the QDS of the conditions when the glass was broken. In all of reality, there is only one collection of glass fragments that exactly matches this one, and there is only one set of ambient conditions which occurred when this exact individual glass was reduced to powder. Even allowing for alternate worlds, there is still only one instance of this glass and these ambient conditions in all of reality, and so the closest QDS link from the glass will be to the rest of the fragments from the same batch, and then the ambient conditions the glass experienced when it was crushed.

If we divide the collection of glass fragments in half and separate them geographically, the QDS they share will allow us to use them like a QDS 'walkie talkie'. This is the basic concept employed in many 'remote link' modules sold for radionics devices. The glass in such a case may instead be a quartz specimen and it may be a few large pieces instead of a crushed powder, but the concept is the same.

If you play an acoustic sound while reducing the glass to powder, you capture the QDS sample of the sound in the glass. If you focus your thoughts and emotions into a specific state while working the mortar and pestle, the QDS of you, the archetypes of the thoughts you are thinking, etc are captured on the glass. This is a good method of capturing long sequences of abstract thought in non-verbal form, as using a mortar and pestle is suitable repetitive to be conducive to meditation.

Since the write media is the fracture lines in the glass, the sample is a permanent one, though each chip is a small piece of solid glass. Solid glass is normally a slowly-rewritable QDS media, but the fracture lines constantly write to the glass, each time it is stimulated by outside energy the energy will carry the QDS held in the glass fracture lines, and update the QDS of the solid glass fragments with the QDS stored in the fracture lines. This applies especially with small or very small fragments.

Even if instead of breaking the glass, we just sand the outside of it under appropriate conditions (running apparatus generating sample to be copied, contact with sample to be copied, or focused mental state), the same or similar results will be derived, and the glass will (after a few minutes) write the stored QDS to any water put into it.

If instead of abrading the entire outer surface, we just make a tiny scratch on the bottom of the glass, we will still sample the QDS of the conditions the glass experienced while being scratched, but not always with enough efficiency to be useful. Also, though the bottom – center is a better place to put our tiny scratch for maximum effect, it is also likely to be scratched over by normal wear on the base of the glass, and then it's gone.

Reducing a material to powder in the presence of the QDS to be copied is often the most effective reduction-sampling method, as it affords ample randomness in proportion to material volume, and allows one to mix some of the programmed glass into another material or contain it in a vessel for convenient use.

This method may be applied to nearly any suitable material, but quartz and glass are excellent choices (aside from the OBVIOUS safety concerns of working with powdered glass). Possible adaptations of this method include sanding, hammer finishing of metal surfaces, distressing of sheet materials, crumpling of sheet materials, chemical reduction, precipitation or deposition processes, etc.

Metamorphic Sampling through Annealing

Let us borrow another example from known physics and look at the (relatively unused / unknown in North America and Europe but apparently popular in Japan) minidisk audio disk format. This is not a business card CD, though it is similar in size and general appearance. A business card CD, like a regular CD, is optical write / optical read media.

A Minidisk is magneto-optical write / optical read media. When the disk is written, a laser is used to heat the metal of the minidisk magneto-optical disk up to the Curie temperature, and then a magnetic field is applied via a magnetic recording head. When the disk is later played, no magnetic read head exists in the player, but the magnetic data on the metallic layer causes a change to the way that light is reflected from the surface of the metal, and the data is gotten off the disk by looking for these characteristic changes to the reflected light (polarity changes I think, but don't quote me on that).

Ordinarily, the metal would not be magnetically responsive enough to be written to by the magnetic recording head in the magneto-optical disk recorder. But when the metal is heated sufficiently by the laser, it changes into a state where it is magnetically responsive enough to be written magnetically, and then allowed to cool so that it will no longer be as susceptible to external magnetic fields, making the sample permanent (in a perfect world). You do not even need another magnetic head to read it; you only need to look at how the magnetic field changes the optical (or other) characteristics of the disk.

What is of importance here for QDS sampling is the fact that when a material goes through a metamorphosis, it changes in terms of what its characteristics are, and that this is an excellent opportunity to impress QDS data on it.

When a material is annealed, either by heating it either to the melting point, or to just below the melting point, or by working it (if it is a flexible material), the QDS data in the material, (all but the raw data describing the material itself) is more or less 'wiped'.

If the material is cold-annealed, then the previous QDS data describing the ambient conditions when the metal was formed and the metals experiences up to this point is erased by blurring it out, as if it were written in a shallow layer of sand and we shook the table and smoothed out all the sand. The previous data is still there but is removed by one order of random variance magnitude, made into a much 'quieter' signal.

If this cold-annealing is done repeatedly, the previous data is pretty much lost but technically still exists. Cold-annealing is useful for both 'clearing' wire and other materials before use in construction of coils, circuits, etc… and for taking small QDS samples of abstract thoughts, audio waves or emf waves.

It is also useful for 'layering' more than one QDS sample onto the same recording media, when it is not convenient produce samples of all the desired entities at the same time or in the same sample format.
Although many complex machines could be built to cold-roll a suitable metal in contact with a QDS sample (or modified to do so) the simplest method is to simply choose a clean, new piece of wire and find a small cup hook in a wall or other vertical surface. Loop the center of the wire over the hook and hold one end of the wire in either hand. Run the wire back and forth over the cup hook a few times along the whole wire length (a few feet) to smooth it out. Then, in the presence of the energy field to be sampled, sharply run a few inches of the wire across the hook with a little tug. Snip out the portion (a few inches is enough) that you just wrote. Keep your mind carefully blank while making the 2 cuts.

The resultant short wire may be placed (appropriately and carefully so as to neither damage the electronics or overheat / repeatedly flex and 'erase' the wire) in contact with conventional electronic / radionic circuits in such a way as to transfer some of the QDS from the wire into the electronics for experimentation. It may also be used as a sample for a radionic machine witness plate or well. It may also be attached to glass in order to program water. The technique may also be adapted to 'program' a coil which will later be used for producing magnetic or scalar fields.

Hot annealing is a more permanent method in that the previous QDS data is more fully erased. If the cold annealing process results in blurring of QDS data (in favor of new data being over-written), then the hot annealing process actually erases the background QDS data and leaves only the raw material data describing the sampling media itself. For example, if the data was chiefly stored in a pattern of metal crystal alignments and microscopic cracks in the metal, then we can realign the molecules by cold rolling and make new cracks holding our new data. But the old cracks still exist, and the 'start point of new molecular alignment will be based on the old QDS held by the metal.

If we melt the metal, we not only realign the metal atoms and given them a new 'start point' for their molecular alignments, but we also melt and reform the metal crystals so that the only data stored in the

cracks between metal crystals will be the 'new' data. Additionally, we make the material temporarily responsive to outside energy fields in a different way than it is when it is solid. Ideally, we would use those energy forms to which the metal is temporarily responsive to write the data.

The material need not be melted, it need only be heated enough to soften and normalize it, then allowed to cool, in the presence of the focused though, sound, EM wave, light wave, picture, scalar field, or other sample to be copied. Recently Phil Ledoux circulated a paper describing how another researcher was heating up a marble with a small torch and playing a solfeggio frequency while it cured. This is an excellent correlation to and example of the dynamics discussed here.

Metamorphic Sampling though chemical process
This is a vast heading with many more applications than will be discussed here; this paper is long enough already. I will mention but two examples here: plastic resin setting and salt crystal growth.
While some materials are such that they can be melted down and resolidified many times (like metal, or some plastics, or water), other materials go through metamorphic processes that are truly irreversible in that once solidified, they cannot be melted down again. Using chemical processes allows us to work with these types of materials, like the various plastic casting resins used to make orgone matrix material. While the addition of metallic elements to the plastic resin does a whole lot of interesting things which are discussed elsewhere, the plastic alone is sufficient to store a QDS concerning the events it experienced while solidifying. In polyester resin, for example, we have a material which first forms small molecules, and then the small molecules form long string shaped molecules, and then the strings curl up and tangle together. The QDS data is stored both on the molecular alignments and on the microscopic cracks that form when the material heats, shrinks, and cools. Adding many small particles of metal produces yet more possible alignments of random bits for the QDS to impress itself on. Once the plastic has cured, you cannot melt it down, it does not melt, and it catches fire. If you burn it, you erase the data. If you dissolve it with acid, you erase the data and / or put the data into the acid.
There are other plastics which smell less than polyester, and they will all work to store QDS samples.

Another method is to set up a chemical process which grows crystals and grow them in the presence of the QDS to be copied. A simple method of doing this is to grow salt or sugar crystals in the presence of the QDS sample. The QDS data will be in the xtal and in the arrangement of many small xtals on the growing surface. If you want to keep the xtals on the growing surface, keep them small and densely packed, when they are grown seal them with a plastic seal coat. Otherwise, let them grow as large as they will grow and take care not to fracture them excessively. Dissolve it in water and the QDS goes into the water.

Methods of reading and broadcasting QDS signals.

Read by contact
Recall the magneto-optical disk and how it reads data off the disk. The changes to reflected light produced by the magnetic component of the disk are subtle changes compared to the way a normal CD player reads a disk, which is why a normal CD player cannot play a magneto-optical disk. Likewise, a normal LP vinyl disk record player will not play back the sounds of ancient Babylon from a pottery shard.

But, due to the effects of contact sampling, a person who has developed 'savant' abilities can gain accurate impressions of the surrounding when both the pottery shard and the LP record were created, by reading the QDS data. Person who do not have developed 'savant' abilities demonstrate biophysical / psychological responses to QDS data reminiscent of hypnotic suggestion or intuitive perception in some cases and reminiscent of conventional exposure to the material source of the recorded QDS data in others, depending on the QDS structure to which the persons are exposed.
This technique may be also used to treat water in pipes if the sample is placed in contact with the exterior of the pipes. Rooms and outdoor areas with QDS encoded paint or paving may be used for contact reading of QDS samples. Drinking and eating vessels similarly treated may be used for contact sampling, or the sample may be used by simply having the person hold it between their hands. If the sample is polarized, then the pos. / output end should go in the left palm and the neg. / input end in the right palm.

Read by throughput
Read by throughput is also a very common method of broadcasting QDS signals. It places a QDS sample in contact with virtually any conventional energy broadcast setup and the QDS is transferred to varying degree on the energy broadcast. Let us

assume that we have the latest orgone matrix material – based QDS sample, which would be something like a small disk or ring shaped casting of EXD orgone matrix material programmed as above. We can shine a light through the hole in the center, if it is a ring.

We can place it on a speaker, and play sound though the speaker to stimulate it. We can place it in contact with or in close proximity to an electrical conductor. These are all methods of getting the QDS into a circuit which is 'broadcasting' something.

If we transfer the sample to an electromagnetic format, then we can broadcast it in a variety of ways using off the shelf audio, video, and other components. These practices are more common in the early 21st century than commonly acknowledged, and therefore I feel comfortable writing about them in the interest of equalizing the playing field between multinational corporate and governmental QDS broadcasts for marketing, political or other purposes and the average person.
While the necessary setup can certainly be enhanced by scalar, orgone, and other technologies dealing with advanced physics, the setup can be functionally simple as a modified audio or video signal, or a specially treated paint applied to an area, or a sample placed strategically in an area near a light, speaker, fan, or other similar energy-circulating device.

It is advantageous to convert the sample into an electromagnetic format since this allows the embedding of it in signals sent to video, audio, radio, or scalar wave, radionic and psionic devices.
Depending on the nature of the QDS to be broadcast, it is often necessary to make an initial, non-permanent sample and then convert this sample to a permanent (non-rewritable) sample. The type of sampling method is often determined by the nature of the sample. For the QDS of a simple sound or frequency or video image, the sound or video image may be directed towards the media while any metamorphic sampling method is used.
But if the QDS is of a long, abstract series of thoughts, then it may to be convenient to first use one of the other methods, and then convert it into an electronic or audio format.

Assuming that convention subliminal encoding methods are not used, the QDS of a complex thought may be partially captured by focusing the thoughts clearly while recording nearly any electronic file or tape.

Mixing it into either white noise or silence increases the amount of the QDS captured.

The subject may or may not be in contact with a suitably placed conductor in the circuit.

A simple setup would be to have the subject touch one of the microphone leads (taking care not to damage the recording apparatus) and focus their thoughts while recording the results. Ideally, this signal should be mixed into something like white noise to give it something to interfere with.

A slightly better setup would be to have the L and R channels of a stereo audio line connected to 2 mobius coils. Have the subject hold both coils, one in each hand or place one coil over each temple. Route low volume white noise through the coils while the subject focuses their thoughts, and send the result to the line in of the audio recorder.

This process may omit the mobius coils, in favor of simpler spiral coils. Mobius coils should not be worn over the temples for prolonged periods, nor powered at more than line levels when used in such a fashion.

The resultant white noise signal will carry the QDS signal. In place of white noise, we can use silence, either a blank carrier signal may be routed through the pickups or each pickup may terminate across suitable resistance. Care should be taken to minimize the 60 hum when recording in this method.

An even better method is to use pump wave pairs (2 signals canceling to produce "loud silence") as the signal routed through the coils. The resultant signal may be mixed into any suitable broadcast system and the QDS will be carried on the audio signal in efficient fashion. In place of a person holding the pickups, the pickups may be placed on either side of a sample of any of the types described above.
This concept may be adapted to recording in optical and magnetic media as well as electronic media.
Radionics devices, psionics devices, orgone accumulators, and other related technologies may be used to improve the efficiency of the basic concepts described here, but that is the subject of another article.

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